

Shayan Shakeri

shayansnezhad@gmail.com | (416) 471-6651 | GitHub | merlyn-labs.com

Experience

AI Researcher & Co-Founder, Merlyn Labs – Toronto Aug 2025 – Present

- Showed $\pi 0.5$ generalization failures on LIBERO-PRO are recipe-induced and solvable; paper under review at CoRL 2026
- Conservative finetuning doubled position-swap success (**21% → 42%**), showing spatial generalization over memorization
- Developing VLM judges that score rollouts into dense, context-dependent RL rewards that are difficult to game
- Open-sourced flow-matching VLA integration for RLinf, enabling RL training on the BEHAVIOR-1K suite in OmniGibson
- Running sim-to-real transfer of household tasks on hand-built AlohaMini embodiment

AI Research Engineer, BMO AI Centre of Excellence – Toronto Sep 2024 – Present

- Uncovered systematic bias to downplay investment risk in a GenAI tool serving \$200B+ AUM wealth management
- Built a deterministic agent eval pipeline using hundreds of synthesized inputs to detect misaligned outputs at scale
- Building a graph-based agentic system answering complex multi-hop relational queries across bank client data
- Leading repair and programming of BMO's physical robot platform

Biomedical & Electrical Engineering Co-op, Epineuron Technologies – Mississauga May 2021 – Aug 2022

- Helped develop PeriPulse, an FDA Breakthrough-designated device now in multi-national clinical trials
- Designed and assembled **PCBs** and authored validation protocols (IEC 60601-1, ISO 13485) for clinical test runs
- Optimized power draw via oscilloscope/power analyzer testing, achieving **900%** battery life improvement
- Modeled electromagnetic field penetration into peripheral nerves in **COMSOL**; data used to optimize electrode diameter

Research & Projects

Stanford BEHAVIOR-1K Challenge – 8th Place, Standard Track Sep 2025 – Nov 2025

- Identified proprioceptive collapse as critical VLA failure mode; 60% masking improved task success by up to **48%**
- Found chunked execution outperformed temporal ensembling by **3x**, revealing VLA architectures lack temporal awareness
- Doubled manipulation success on long-tail subtasks by oversampling skill transitions via boundary resampling
- Trained on 10,000+ demonstrations across 22 tasks; published a technical report and further analysis on LessWrong

BardSong, AI-Powered D&D Storytelling Tool Jun 2025 – Present

- Built end-to-end pipeline: Groq speech-to-text → Gemini scene extraction → image generation → narrated video
- Fine-tuned LLM on D&D sourcebooks for improved narrative coherence; currently in closed alpha with **23 DMs**

Voice-Controlled Robotic Prosthetic, University of Toronto Sep 2024 – Mar 2025

- Developed LLM-based control pipeline converting natural language to manipulation sequences for 7-DOF simulated arm
- Integrated YOLO object detection with LiDAR depth mapping for 3D localization and grasp planning

VP Technical, McMaster Medical Design Team Apr 2018 – Apr 2023

- Led development of open-source motorized prosthetic for lower-arm amputees that anyone could assemble for **<\$300**

Education

M.Eng, MIE – AI & Robotics, University of Toronto (Anticipated Graduation: 04/27) Sep 2024 – Apr 2027

Coursework: Deep Learning, Reinforcement Learning, AI Applications in Robotics, Healthcare Robotics

B.Eng, Engineering Physics Co-op, McMaster University – Dean's Honour List Sep 2018 – Jun 2023

Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems

Skills & Interests

ML & AI: Python, C/C++, PyTorch, JAX, RL, Imitation Learning, LLM Fine-tuning, Alignment Evaluation, Agents

Robotics & Hardware: VLA Models, OmniGibson, MuJoCo, Flow Matching, Sim2Real, ROS2, PCB Design, 3D Printing

Interests: Literature, Dungeons & Dragons, Volleyball, Twilight Imperium, Piano